

Internship in Physical Organic Chemistry

Week 2 — Part 1: Lab Practice and Data Management

Prepared by Dr. Alexandra Tsybizova

Goals of this part:

- Understand what a lab journal is and how to fill it correctly
- Set up a logical folder structure and file naming convention for your data
- Learn to produce publication-quality spectra from exported data
- Set up the Week 2 sections of your `scratch.tex` working log

This part introduces the practical habits that underpin all experimental work. Before you record a single spectrum, you must know how to document what you do, where to store what you produce, and how to present it. These tasks are part of doing science correctly. Every result you produce must be properly documented. A result that cannot be traced back to documented conditions is not a result.

1 The lab journal

A lab journal is the primary record of your experimental work. It is a contemporaneous record: you write in it *while* you work.

The lab journal has a specific function that no other document can replace. If a result is questioned — by your supervisor, by a reviewer, or by yourself six months later — the lab journal is the only place where the full experimental context is recorded. It answers the questions: what did I actually do, exactly, when did I do it, and what did I see?

1.1 What belongs in a lab journal entry

Every experimental session produces one lab journal entry. A complete entry contains the following elements, always in this order:

1. **Date and name of the experimenter.** Written at the top of every entry without exception.
2. **Objective.** One or two sentences stating what you intend to do and why. This is written *before* you start. Example: *Record positive-ion ESI-MS and MS/MS spectra of compound X in methanol at 10^{-5} M to identify the molecular ion and major fragment ions.*
3. **Procedure as executed.** A step-by-step account of what you actually did, written as you do it, documenting what happened. If you deviated from your initial plan, that deviation must be recorded. Include quantities, concentrations, instrument settings as set (not as intended), and the names of the raw data files you saved.
4. **Observations.** What you saw, measured, or noted during the experiment. Observations are direct: *the molecular ion appears at m/z XXX* is an observation. *The compound fragmented as expected* is not — it mixes observation with interpretation. Write observations as they occur. Do not wait until the end of the session.
5. **Preliminary conclusions and next steps.** Written at the end of the session. What do the observations suggest? What needs to be done next? This section may be brief, but it must be present.

The distinction between observation and interpretation that you practised in Week 1 when reading papers applies directly here. In your lab journal, keep observations and interpretations clearly separated. An observation is what the instrument reported. An interpretation is what you think it means. Both are valuable, but they must not be mixed.

1.2 How to fill the lab journal template

A standalone lab journal template is provided (`labjournal_template.pdf`). Print one copy per experimental session, or fill it in on your laptop. You will use it on Day 1 (instrument training) and on each acquisition day.

Use the template as a guide: If you need more space for observations, use the back of the page or attach a separate sheet.

After each session, photograph or scan your completed journal pages and place them in the `labjournal/` folder described in Section 2. Your supervisor reads the lab journal to follow your progress between sessions.

2 Data management

Every file you produce during the experiment must be named unambiguously and stored in a predictable location. A spectrum with an unclear name or stored in the wrong folder cannot be found, cannot be cited, and cannot be reproduced.

2.1 Folder structure

Create the following folder structure on your computer at the start of Week 2. Use exactly these names — your partner uses the same structure, which makes sharing and comparison straightforward.

```
week02/  
  raw/  
  exported/  
  figures/  
  labjournal/
```

- `raw/` — all `.raw` files directly from Xcalibur, unmodified
- `exported/` — all `.xlsx` files exported from Xcalibur
- `figures/` — final figures saved from Excel as `.pdf` or `.png`
- `labjournal/` — scanned or photographed lab journal pages (`.pdf` or `.jpg`)

Never move, rename, or modify files in `raw/` after saving them there. Raw files are your primary data. Everything else is derived from them.

2.2 File naming convention

Every file must be named according to the following convention:

$$[\text{date}]_ [\text{compound}]_ [\text{ionmode}]_ [\text{scantype}]_ [\text{details}]$$

where:

- `[date]` — YYYYMMDD format (always first, enables automatic chronological sorting)
- `[compound]` — abbreviated compound name, lowercase, no spaces
- `[ionmode]` — `pos` or `neg`

- [scantype] — MS1 or MS2
- [details] — for MS2 files: precursor m/z and NCE value; omit for MS1 files

Example file names for this project:

20250601_luteolin_pos_MS1.raw
 20250601_luteolin_pos_MS2_precursor287_NCE30.raw
 20250601_luteolin_neg_MS2_precursor285_NCE25.raw
 20250601_quercetin_pos_MS1.raw
 20250601_quercetin_neg_MS2_precursor301_NCE30.raw

The exported Excel files follow the same names with .xlsx extension:

20250601_luteolin_pos_MS1.xlsx

Apply the naming convention to every file, including files you consider preliminary or uncertain. The name **test.raw** or **spectrum1.raw** is never acceptable. If you are unsure whether a spectrum is useful, save it with the correct name anyway and decide later whether to use it. You cannot rename a raw file without breaking its traceability.

2.3 Backup

At the end of each working day, copy your entire **week02/** folder to a second location: either cloud storage (your university OneDrive, Google Drive, or equivalent) or a USB drive. Raw instrument data cannot be recovered if lost. Record the backup in your lab journal entry for that day.

3 Viewing and exporting data from Xcalibur

All acquired spectra are stored as **.raw** files and viewed in **Xcalibur**. To open Xcalibur, launch it from the instrument computer desktop. The main screen is the Roadmap (Figure 1), which gives access to all software modules. To view your spectra, click **Qual Browser**.

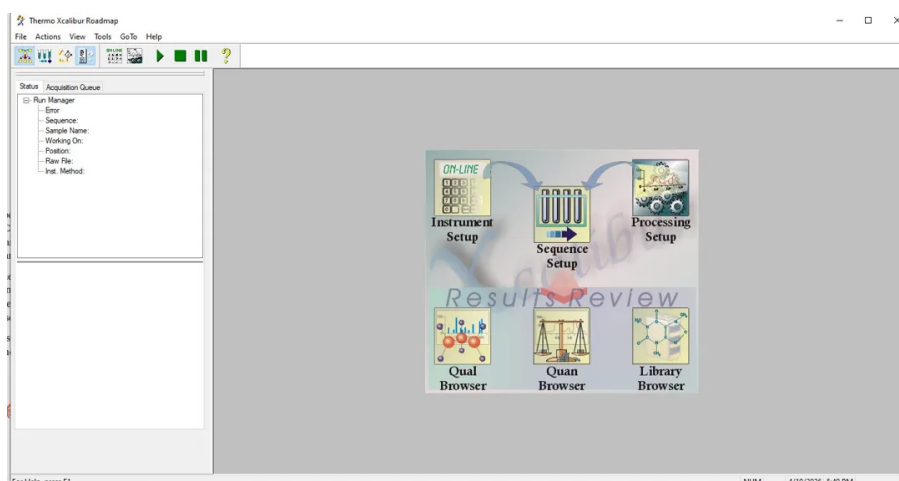


Figure 1: The Thermo Xcalibur Roadmap. To view acquired spectra, open the **Qual Browser**.

In the Qual Browser, open your **.raw** file using File → Open. The interface shows two panels (Figure 2). The upper panel is the total ion chromatogram (TIC), which shows total ion signal over time. For direct infusion experiments, this will appear as a roughly flat trace. The lower

panel shows the mass spectrum for the scan or range of scans currently selected. Click on a point in the TIC to display the corresponding spectrum, or drag to select and average a range of scans. The scan header above the lower panel shows the file name, scan number, retention time, and acquisition parameters — record these in your lab journal.

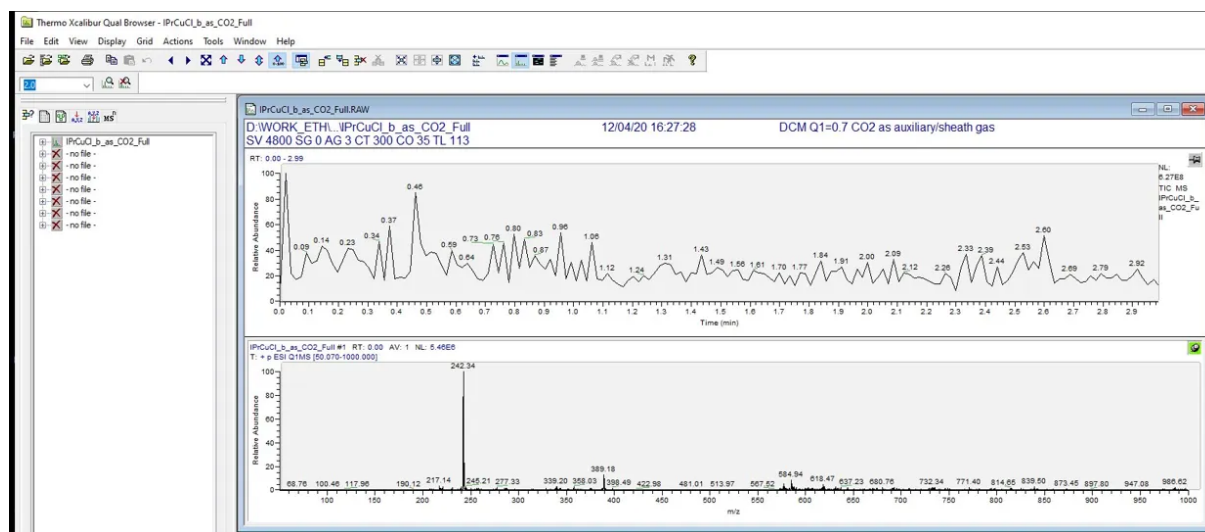


Figure 2: The Qual Browser interface. The upper panel shows the total ion chromatogram (TIC); the lower panel shows the mass spectrum for the selected scan. The scan header shows the file name, scan number, retention time, and acquisition parameters.

To export a spectrum to Excel, right-click on the lower spectrum panel and select Export → Clipboard (Exact Mass) (Figure 3). This copies the complete peak list to the clipboard. Open Excel, create a new sheet, and paste (Ctrl+V).

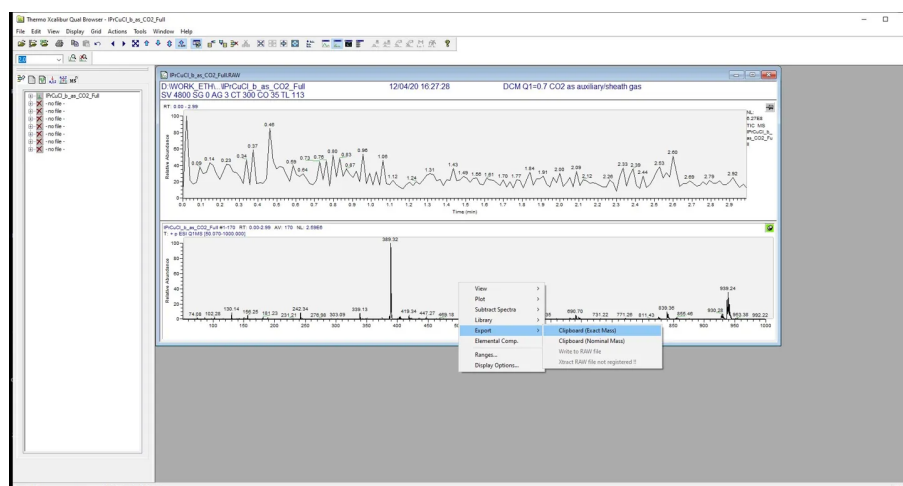


Figure 3: Exporting a spectrum from Qual Browser. Right-click on the spectrum panel and select Export → Clipboard (Exact Mass). The data can then be pasted directly into Excel.

Save the Excel file immediately using the file naming convention from Section 2, with **.xlsx** extension, into your **week02/exported/** folder.

4 Producing figures from exported data

A spectrum exported from Xcalibur to Excel contains profile data: two columns of mass and intensity values with many closely spaced points across the full scan range. Before this data can appear in your report, it must be normalised, plotted, and annotated.

4.1 Preparing the data

When you paste the clipboard export into Excel, the first seven rows contain metadata (file name, scan type, scan number, RT, AV, data points). Delete these rows. Row 8 contains the column headers Mass and Intensity — keep it. The data starts from row 9 (Figure 4).

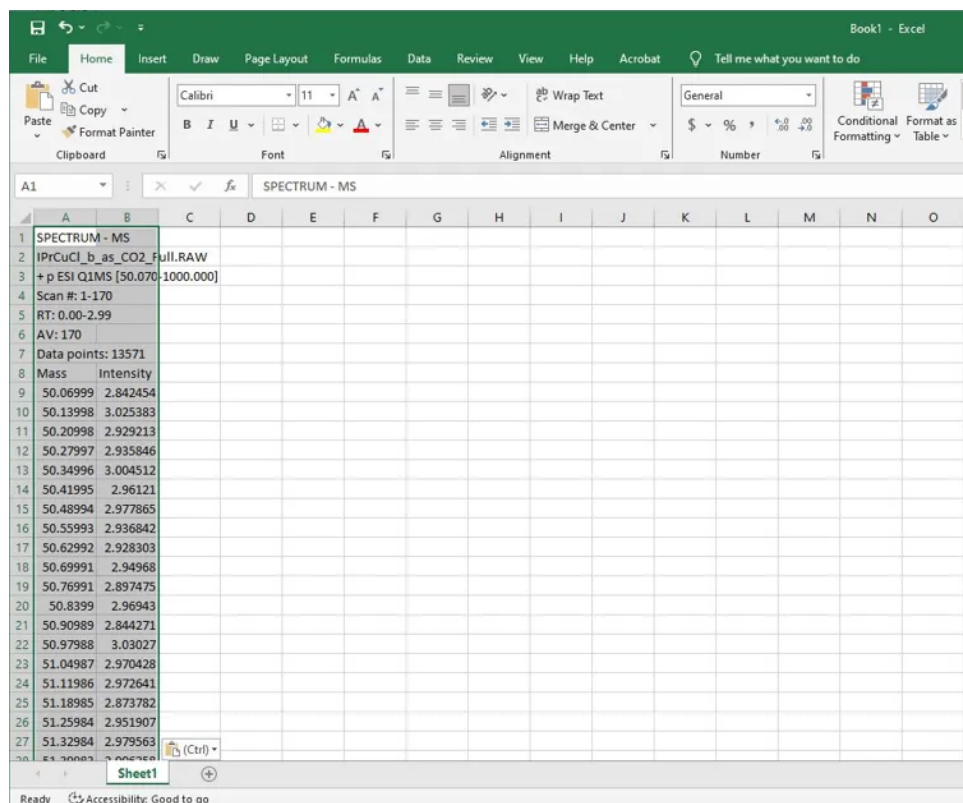


Figure 4: Exported spectrum data pasted into Excel. Rows 1–7 contain metadata and must be deleted before plotting. The data columns (Mass and Intensity) begin at row 8.

All mass spectra in scientific publications are displayed as *relative intensity*, normalised to the most abundant peak (base peak = 100%). To normalise:

1. Find the maximum intensity: in an empty cell, enter `=MAX(B:B)` and note the cell reference.
2. Add a third column labelled **Relative intensity (%)**.
3. In the first data cell of this column, enter `=(B9/MAX($B:$B))*100`. Adjust the row number to match your layout.
4. Copy the formula down for all data rows.

4.2 Building the chart

1. Select the Mass column and the Relative intensity (%) column together.
2. Insert a **Line chart**. The exported data is profile data with many closely spaced points. A line chart connects these points and naturally produces the correct spectrum appearance — sharp peaks rising from a flat baseline. Do not use a Column or Bar chart here.

4.3 Formatting the chart

A publication-quality spectrum has the following properties. Apply each one explicitly — Excel's default chart style is not acceptable.

- ***x*-axis label:** m/z (italicised in the report; plain text in Excel is acceptable)
- ***y*-axis label:** Relative intensity (%)
- ***y*-axis range:** 0 to 100, with no automatic scaling beyond 100
- ***x*-axis range:** set the minimum to a value slightly below your lowest peak, not to zero (a spectrum starting at m/z 0 wastes most of the figure space)
- **No chart title** — the figure caption in your report serves this function
- **No legend** — there is only one data series
- **No gridlines**
- **No coloured background** — white only
- **Line colour:** black

4.4 Annotating peaks

You do not need to label every peak. Label the peaks you will discuss in your report: the molecular ion (or the most abundant ion if no molecular ion is visible), major fragments, and any peak whose identity you have assigned.

To annotate a peak, use a text box positioned above the bar. Write the m/z value to one decimal place (Orbitrap data has sufficient resolution to report fractional m/z values meaningfully).

Which peaks to annotate for this project:

At minimum, annotate the precursor ion, the base peak, and any fragment ion you assign to an RDA pathway. Additional annotations may be added once you complete the structure characterisation step in Part 3.

4.5 Exporting the figure

Save the final figure as a `.pdf` file (File → Save As → PDF) or as a `.png` at 300 dpi minimum. Place it in `week02/figures/` using the same naming convention as the source file, with `_fig` appended:

20250601_luteolin_pos_MS1_fig.pdf

To insert the figure into your Overleaf report, upload the file to your Overleaf project and use the standard L^AT_EX figure environment.

5 Working log

As in Week 1, you will maintain a working log that captures all intermediate thinking produced this week. The Week 2 working log is provided as a standalone file (`scratch_week02.tex`). Add `\input{scratch_week02}` to your Overleaf main file, immediately after `\input{scratch}`.

Each section of the working log is referenced from the part document where it is relevant: Section 2W-a is introduced in Part 2 (sample preparation), Sections 2W-b through 2W-d in Part 3 (acquisition and structure characterisation), and Sections 2W-e and 2W-f in Part 4 (writing the Experimental chapter).

Write complete sentences in every section of the working log — not keywords or fragments. Your supervisor reads it as a compiled PDF to follow your progress between sessions.